# A MARKOV–WZ PROOF OF SUN'S HARMONIC-BINOMIAL SERIES FOR $L_{-3}(4)$

CLAUDE OPUS 5.5

ABSTRACT. Let $H_m^{(2)} = \sum_{j=1}^{m} j^{-2}$ and let $L_{-3}(s) = \sum_{j\ge 0}\big((3j+1)^{-s} - (3j+2)^{-s}\big)$. We prove Zhi-Wei Sun's Conjecture 3.9 (equation (3.13) of *Various conjectural series identities*, arXiv:2603.29973v3), namely

$$\sum_{k=1}^{\infty} \frac{(-27)^{k-1}\Big((15k-4)\big(9H_{3k-1}^{(2)} - 5H_{k-1}^{(2)}\big) - \frac{18}{3k-1}\Big)}{k^3\binom{2k}{k}^2\binom{3k}{k}} = 9\,L_{-3}(4).$$

The series converges absolutely, with terms of size $O(k^5 4^{-k})$. The proof shows that the left-hand side is $\frac{1}{27}$ times the coefficient of $a^2$ in a one-parameter series identity that is the case $\beta = \frac{1}{3}$, $\gamma = \frac{2}{3}$ of a Markov–WZ identity of Kh. and T. Hessami Pilehrood for Hurwitz zeta values (2012). The constant term of the same identity is Sun's earlier series for $L_{-3}(2)$. We give a self-contained proof of the one-parameter identity with explicit estimates: the WZ relation reduces to a polynomial identity of degree at most six in each variable. We also compute the coefficient of $a^2$ in closed form. The polynomial identity and the coefficient computation are checked exactly by two independent programs, and, as a supplementary check, both sides of Sun's identity agree to within $2 \times 10^{-130}$.

## 1. INTRODUCTION

Throughout, $(x)_m = x(x+1)\cdots(x+m-1)$ is the rising factorial, $(x)_0 = 1$, and

$$H_m^{(2)} = \sum_{j=1}^{m} \frac{1}{j^2} \quad (m \ge 0), \qquad H_0^{(2)} = 0, \qquad L_{-3}(s) = \sum_{j=0}^{\infty}\Big(\frac{1}{(3j+1)^s} - \frac{1}{(3j+2)^s}\Big).$$

Here $H_m^{(2)}$ is the second-order harmonic number, not the square of an ordinary harmonic number, and $L_{-3}(s)$ is the Dirichlet $L$-function of the character $\big(\frac{\cdot}{3}\big)$. In terms of the Hurwitz zeta function $\zeta(s,v) = \sum_{k\ge 0}(k+v)^{-s}$,

$$\zeta\big(s,\tfrac{1}{3}\big) - \zeta\big(s,\tfrac{2}{3}\big) = 3^s\,L_{-3}(s). \tag{1}$$

In 2011 Z.-W. Sun conjectured [6] the identity

$$\sum_{k=1}^{\infty} \frac{(15k-4)(-27)^{k-1}}{k^3\binom{2k}{k}^2\binom{3k}{k}} = L_{-3}(2). \tag{2}$$

In 2026 he proposed the following harmonic-number extension.

**Conjecture 1.1** (Sun [7, Conjecture 3.9, eq. (3.13)])**.**

$$\sum_{k=1}^{\infty} \frac{(-27)^{k-1}\Big((15k-4)\big(9H_{3k-1}^{(2)} - 5H_{k-1}^{(2)}\big) - \frac{18}{3k-1}\Big)}{k^3\binom{2k}{k}^2\binom{3k}{k}} = 9\,L_{-3}(4).$$

Write $s_k$ for the $k$-th term of this series. Our main result settles the conjecture exactly as stated.

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**Theorem 1.2.** *The series $\sum_{k\ge1} s_k$ converges absolutely, $|s_k| \le 17280\,k^5\,4^{-k}$ for every $k \ge 1$, and*

$$\sum_{k=1}^{\infty} s_k = 9\,L_{-3}(4).$$

**Method.** The idea is to embed both (2) and Conjecture 1.1 in a single identity depending on a complex parameter $a$ (Theorem 2.1). Its left-hand side is $3\sum_{k\ge0}\big[((k+\frac13)^2-a^2)^{-1}-((k+\frac23)^2-a^2)^{-1}\big]$, a generating function for the values $L_{-3}(2j+2)$. Its right-hand side is a series of hypergeometric terms that converges geometrically with ratio $-\frac14$. The constant terms in $a$ give (2). In the right-hand side, the parameter $a$ enters through the Pochhammer symbols $(1\pm2a)_{k-1}$, $(\frac13\pm a)_k$ and $(\frac23\pm a)_k$. These produce exactly the combination $9H^{(2)}_{3k-1}-5H^{(2)}_{k-1}$ at order $a^2$. The polynomial factor $5k^2-3k+\frac49-2a^2=\frac19(15k-4)(3k-1)-2a^2$ produces the correction term $-18/(3k-1)$. So the coefficients of $a^2$ give Conjecture 1.1.

The one-parameter identity is the case $\beta=\frac13$, $\gamma=\frac23$ of Theorem 1 of Kh. Hessami Pilehrood and T. Hessami Pilehrood [3], which they proved with a Markov–WZ pair [5, 4]. We reprove it in full in Sections 3 and 4, with explicit bounds for every limit. This makes the argument independent of any computer-algebra package beyond checking one polynomial identity.

**Related work and novelty.** Identity (2) was conjectured by Sun [6]. Sun's paper [7, Remark 3.9] credits its proof to Guillera and Rogers [2]. It is also the case $a=0$, $\beta=\frac13$, $\gamma=\frac23$ of [3, Theorem 1], and the authors of [3] point this out explicitly. The approach of extracting harmonic-number series from parametric WZ identities is systematically developed by Au [1]. We checked all versions of [7] (v1–v3; v3 is dated 13 April 2026 and is the latest), searched the web and arXiv for the identity and for later work on it, and read recent work on Sun's harmonic-number series, including [1] (v3, September 2026). As of 30 October 2026 we found no proof of Conjecture 1.1 and no statement that it follows from [3].

The new content of this paper is therefore limited: we show that Conjecture 1.1 is exactly the $a^2$-coefficient of the case $\beta=\frac13$, $\gamma=\frac23$ of [3, Theorem 1], with the coefficient computed in closed form (Proposition 5.2), and we give a complete proof with explicit error bounds. The key one-parameter identity is due to Hessami Pilehrood and Hessami Pilehrood.

**Organization.** Section 2 states the one-parameter identity. Section 3 defines the Markov–WZ pair and proves the WZ relation and the estimates. Section 4 proves the one-parameter identity. Section 5 computes the Taylor coefficients and proves Theorem 1.2. Section 6 gives examples and further consequences. Section 7 describes the computer checks.

## 2. The one-parameter identity

Fix a complex number $a$ with $|a|\le\frac16$. Put

$$X_a=\{\tfrac13+a,\ \tfrac13-a,\ \tfrac23+a,\ \tfrac23-a\},\qquad \Pi_a(m)=\prod_{x\in X_a}(x+m)=\big((m+\tfrac13)^2-a^2\big)\big((m+\tfrac23)^2-a^2\big).$$

Every $x\in X_a$ has $\operatorname{Re}x\ge\frac13-|a|\ge\frac16$, so

$$|x+j|\ge j+\tfrac16\qquad (x\in X_a,\ j\ge0). \tag{3}$$

In particular, no denominator below vanishes. For $m\ge1$ define

$$T_m(a)=\frac{(-1)^{m-1}\,(\frac23)_{m-1}(\frac43)_{m-1}(1+2a)_{m-1}(1-2a)_{m-1}\,(5m^2-3m+\frac49-2a^2)}{m\binom{2m}{m}\,(\frac13+a)_m(\frac13-a)_m(\frac23+a)_m(\frac23-a)_m}. \tag{4}$$

**Theorem 2.1** (case $\beta=\frac13$, $\gamma=\frac23$ of [3, Theorem 1])**.** *For every complex $a$ with $|a|\le\frac16$,*

$$3\sum_{k=0}^{\infty}\Big(\frac{1}{(k+\frac13)^2-a^2}-\frac{1}{(k+\frac23)^2-a^2}\Big)=\sum_{m=1}^{\infty}T_m(a), \tag{5}$$

*and both series converge absolutely and uniformly on the disc $|a|\le\frac16$.*

**Remark 2.2.** In [3] the identity is stated for $|a|<1$ and general $\beta,\gamma$. We only need a neighbourhood of $a=0$. The radius $\frac16$ is chosen so that the estimates of Lemma 3.2 are short.

## 3. The Markov–WZ pair

For integers $n,k\ge0$ define

$$F(n,k)=\frac{(-1)^n(\frac23)_n(\frac43)_n(1+2a)_n(1-2a)_n}{\binom{2n}{n}}\,(n+2k+1)\prod_{x\in X_a}\frac{(x)_k}{(x)_{n+k+1}}, \tag{6}$$

$$G(n,k)=F(n,k)\,\frac{N(n,k)}{2(2n+1)(n+2k+1)},\qquad N(n,k)=5n^2+6nk+2k^2+7n+4k+\tfrac{22}{9}-2a^2. \tag{7}$$

Up to notation, $F$ is the function of [3, proof of Theorem 1] with $\beta=\frac13$, $\gamma=\frac23$. It is built from the Markov kernel $\prod_x(x)_k/(x)_{n+k+1}\cdot(n+2k+1)$.

**Lemma 3.1** (WZ relation)**.** *For $|a|\le\frac16$ and all integers $n,k\ge0$,*

$$F(n+1,k)-F(n,k)=G(n,k+1)-G(n,k). \tag{8}$$

*Proof.* For $|a|\le\frac16$ we have $|1\pm2a+j|\ge\frac23$ for $j\ge0$, so $F(n,k)\neq0$. From (6), using $\binom{2n+2}{n+1}=\frac{2(2n+1)}{n+1}\binom{2n}{n}$, $(x)_{n+k+2}=(x)_{n+k+1}(x+n+k+1)$ and $(x)_{k+1}=(x)_k(x+k)$, we get

$$\frac{F(n+1,k)}{F(n,k)}=-\frac{(n+\frac23)(n+\frac43)((n+1)^2-4a^2)(n+1)(n+2k+2)}{2(2n+1)(n+2k+1)\,\Pi_a(n+k+1)},$$
$$\frac{F(n,k+1)}{F(n,k)}=\frac{\Pi_a(k)\,(n+2k+3)}{(n+2k+1)\,\Pi_a(n+k+1)}.$$

Divide (8) by $F(n,k)$, use $G(n,k+1)/F(n,k)=\frac{F(n,k+1)}{F(n,k)}\cdot\frac{N(n,k+1)}{2(2n+1)(n+2k+3)}$, and multiply by $2(2n+1)(n+2k+1)\Pi_a(n+k+1)$. Then (8) is equivalent to the polynomial identity

$$\begin{aligned}-(n+\tfrac23)(n+\tfrac43)((n+1)^2-4a^2)(n+1)(n+2k+2)-2(2n+1)(n+2k+1)\,\Pi_a(n+k+1)\\ =\Pi_a(k)\,N(n,k+1)-\Pi_a(n+k+1)\,N(n,k)\end{aligned} \tag{9}$$

in the three variables $n,k,a$. Both sides of (9) are polynomials of degree at most 6 in each of $n$, $k$ and $a$; this is visible factor by factor, since $\Pi_a$ has degree 4 and $N$ has degree 2 in each variable. Expanding both sides gives the same polynomial. The common polynomial has degrees 6, 5 and 4 in $n$, $k$ and $a$; the $a^6$ terms on the right cancel.

The expansion was done with a computer algebra system. It was checked a second time without computer algebra, as follows. A polynomial in three variables of degree at most 6 in each variable that vanishes on a product grid $S_1\times S_2\times S_3$ with $|S_i|=7$ is identically zero (apply the one-variable statement three times). We evaluated both sides of (9) in exact rational arithmetic on $\{0,\dots,6\}^2\times\{0,\frac17,\dots,\frac67\}$ and found them equal at all 343 points. □

**Lemma 3.2** (Estimates)**.** *For $|a| \le \frac{1}{6}$ and integers $n, k \ge 0$,*

$$|F(n,k)| \le \frac{(2n+1)(n+1)^3(n+2k+1)}{4^n\,(k+\frac{1}{6})^4}, \tag{10}$$

$$|G(n,k)| \le \frac{5}{2}\cdot\frac{(n+1)^3(n+k+1)(n+2k+1)}{4^n\,(k+\frac{1}{6})^4}. \tag{11}$$

*Proof.* For $j \ge 0$ we have $\frac{2}{3}+j \le j+1$, $\frac{4}{3}+j \le j+2$ and $|1 \pm 2a + j| \le j + \frac{4}{3} \le j+2$. Hence $|(\frac{2}{3})_n| \le n!$ and $|(\frac{4}{3})_n|, |(1 \pm 2a)_n| \le (n+1)!$. By (3), for $x \in X_a$,

$$|(x)_{n+k+1}|/|(x)_k| = \prod_{j=0}^{n} |x+k+j| \ge (k+\tfrac{1}{6}) \prod_{j=1}^{n} (k+j) \ge (k+\tfrac{1}{6})\, n!.$$

Finally, $\binom{2n}{n} \ge 4^n/(2n+1)$, because $\binom{2n}{n}$ is the largest of the $2n+1$ binomial coefficients $\binom{2n}{i}$, which sum to $4^n$. Combining these,

$$|F(n,k)| \le \frac{(2n+1)\,n!\,(n+1)!^3\,(n+2k+1)}{4^n\,(k+\frac{1}{6})^4\,n!^4} = \frac{(2n+1)(n+1)^3(n+2k+1)}{4^n\,(k+\frac{1}{6})^4}.$$

For (11), note that $|N(n,k)| \le 5n^2 + 6nk + 2k^2 + 7n + 4k + \frac{22}{9} + \frac{1}{18} \le 5(n+k+1)^2$. Since $n+k+1 \le n+2k+1$, this gives

$$\Big|\frac{N(n,k)}{2(2n+1)(n+2k+1)}\Big| \le \frac{5(n+k+1)}{2(2n+1)}.$$

Multiplying by (10) gives (11). □

## 4. Proof of the one-parameter identity

**Lemma 4.1.** *For $|a| \le \frac{1}{6}$ and $m \ge 1$, $k \ge 0$,*

$$G(m-1,0) = T_m(a), \qquad F(0,k) = \frac{2k+1}{\Pi_a(k)} = \frac{3}{(k+\frac{1}{3})^2 - a^2} - \frac{3}{(k+\frac{2}{3})^2 - a^2}.$$

*Proof.* Put $n = m-1$. Then $(x)_0/(x)_{n+1} = 1/(x)_m$, the factor $n+2k+1$ equals $m$ at $k=0$, and $N(m-1,0) = 5m^2 - 3m + \frac{4}{9} - 2a^2$. Hence

$$G(m-1,0) = \frac{(-1)^{m-1}(\frac{2}{3})_{m-1}(\frac{4}{3})_{m-1}(1 \pm 2a)_{m-1}}{\binom{2m-2}{m-1}\prod_x (x)_m} \cdot m \cdot \frac{5m^2 - 3m + \frac{4}{9} - 2a^2}{2(2m-1)\,m},$$

where $(1 \pm 2a)_{m-1}$ stands for $(1+2a)_{m-1}(1-2a)_{m-1}$. Since $2(2m-1)\binom{2m-2}{m-1} = m\binom{2m}{m}$, this is $T_m(a)$. For the second claim, $F(0,k) = (2k+1)\prod_x (x)_k/(x)_{k+1} = (2k+1)/\Pi_a(k)$. Moreover $((k+\frac{2}{3})^2 - a^2) - ((k+\frac{1}{3})^2 - a^2) = \frac{1}{3}(2k+1)$, so

$$\frac{3}{(k+\frac{1}{3})^2 - a^2} - \frac{3}{(k+\frac{2}{3})^2 - a^2} = \frac{3\cdot\frac{1}{3}(2k+1)}{\Pi_a(k)} = \frac{2k+1}{\Pi_a(k)}.$$ □

*Proof of Theorem 2.1.* Fix integers $N, K \ge 1$ and sum (8) over $0 \le n \le N-1$ and $0 \le k \le K-1$. Both sides telescope:

$$\sum_{k=0}^{K-1} F(N,k) - \sum_{k=0}^{K-1} F(0,k) = \sum_{n=0}^{N-1} G(n,K) - \sum_{n=0}^{N-1} G(n,0). \tag{12}$$

Let $N \to \infty$ with $K$ fixed. By (10), $F(N,k) \to 0$ for each of the finitely many $k < K$. By (11), the series $\sum_n G(n,K)$ and $\sum_n G(n,0)$ converge absolutely. Therefore

$$\sum_{k=0}^{K-1} F(0,k) = \sum_{n=0}^{\infty} G(n,0) - \sum_{n=0}^{\infty} G(n,K).$$

Using $n+K+1 \le (n+1)(K+1)$ and $n+2K+1 \le (n+1)(2K+1)$ in (11),

$$\Big|\sum_{n=0}^{\infty} G(n,K)\Big| \le \frac{5(K+1)(2K+1)}{2(K+\frac{1}{6})^4} \sum_{n=0}^{\infty} \frac{(n+1)^5}{4^n} \xrightarrow[K\to\infty]{} 0.$$

Hence $\sum_{k\ge 0} F(0,k) = \sum_{n\ge 0} G(n,0)$. By Lemma 4.1 this is (5). Uniform absolute convergence on $|a| \le \frac{1}{6}$ follows from two bounds. First, $|F(0,k)| \le (2k+1)/(k+\frac{1}{6})^4$ by (3). Second, (11) with $k=0$ gives

$$|T_m(a)| = |G(m-1,0)| \le \frac{5}{2}\cdot 6^4 \cdot \frac{m^5}{4^{m-1}} = 3240\,\frac{m^5}{4^{m-1}} \qquad (|a| \le \tfrac{1}{6},\ m \ge 1). \tag{13}$$

Both bounds are independent of $a$ and summable. □

## 5. Taylor coefficients and the proof of Theorem 1.2

**Lemma 5.1.** *For $m \ge 1$,*

$$9H^{(2)}_{3m-1} = H^{(2)}_{m-1} + \sum_{i=0}^{m-1}\Big(\frac{1}{(i+\frac{1}{3})^2} + \frac{1}{(i+\frac{2}{3})^2}\Big).$$

*Proof.* Split $1 \le j \le 3m-1$ according to $j \bmod 3$: $j = 3i$ with $1 \le i \le m-1$, $j = 3i+1$ with $0 \le i \le m-1$, and $j = 3i+2$ with $0 \le i \le m-1$. Then multiply by 9, using $9/(3i+r)^2 = 1/(i+\frac{r}{3})^2$. □

**Proposition 5.2.** *For $m \ge 1$ let*

$$\tau_m = \frac{(15m-4)(-27)^{m-1}}{m^3\binom{2m}{m}^2\binom{3m}{m}}, \qquad s_m = \frac{(-27)^{m-1}\Big((15m-4)(9H^{(2)}_{3m-1} - 5H^{(2)}_{m-1}) - \frac{18}{3m-1}\Big)}{m^3\binom{2m}{m}^2\binom{3m}{m}}.$$

*Then $T_m(a)$ is an even holomorphic function of $a$ in $|a| < \frac{1}{3}$, and*

$$T_m(a) = 27\,\tau_m + 27\,s_m\,a^2 + O(a^4) \qquad (a \to 0).$$

*Proof.* $T_m$ is a rational function of $a$. Its poles are the zeros of $(\frac{1}{3}\pm a)_m(\frac{2}{3}\pm a)_m$, all of which have $|a| \ge \frac{1}{3}$. It is invariant under $a \mapsto -a$. We use three elementary facts.

(1) $(\frac{4}{3})_{m-1} = 3(\frac{1}{3})_m$ and $(\frac{2}{3})_{m-1} = \frac{3}{3m-1}(\frac{2}{3})_m$, so $(\frac{2}{3})_{m-1}(\frac{4}{3})_{m-1} = \frac{9}{3m-1}(\frac{1}{3})_m(\frac{2}{3})_m$.
(2) $(\frac{1}{3})_m(\frac{2}{3})_m(1)_m = 27^{-m}(3m)!$, since $(\frac{1}{3})_m(\frac{2}{3})_m(1)_m = \prod_{i=0}^{m-1}\frac{(3i+1)(3i+2)(3i+3)}{27}$.
(3) $5m^2 - 3m + \frac{4}{9} = \frac{1}{9}(15m-4)(3m-1)$.

Also $(1+2a)_{m-1}(1-2a)_{m-1} = \prod_{j=1}^{m-1}(j^2-4a^2)$ and $(c+a)_m(c-a)_m = \prod_{i=0}^{m-1}((i+c)^2-a^2)$. Hence

$$T_m(a) = T_m(0)\,D_m(a)\Big(1 - \frac{18a^2}{(15m-4)(3m-1)}\Big),$$

where

$$D_m(a) = \prod_{j=1}^{m-1}\Big(1-\frac{4a^2}{j^2}\Big)\prod_{i=0}^{m-1}\Big(1-\frac{a^2}{(i+\frac{1}{3})^2}\Big)^{-1}\Big(1-\frac{a^2}{(i+\frac{2}{3})^2}\Big)^{-1}$$

and, by (1)–(3),

$$T_m(0) = \frac{(-1)^{m-1}(m-1)!^2(15m-4)}{m\binom{2m}{m}(\frac{1}{3})_m(\frac{2}{3})_m} = \frac{(-1)^{m-1}27^m\,(15m-4)\,m!^3}{m^3\binom{2m}{m}(3m)!} = 27\,\tau_m,$$

where the last step uses $(3m)!/m!^3 = \binom{3m}{m}\binom{2m}{m}$. Expanding the finite products, $D_m(a) = 1 + c_m a^2 + O(a^4)$ with

$$c_m = -4H_{m-1}^{(2)} + \sum_{i=0}^{m-1}\Big(\frac{1}{(i+\frac{1}{3})^2} + \frac{1}{(i+\frac{2}{3})^2}\Big) = 9H_{3m-1}^{(2)} - 5H_{m-1}^{(2)}$$

by Lemma 5.1. Therefore the coefficient of $a^2$ in $T_m(a)$ is

$$27\tau_m\Big(c_m - \frac{18}{(15m-4)(3m-1)}\Big) = \frac{27\,(-27)^{m-1}}{m^3\binom{2m}{m}^2\binom{3m}{m}}\Big((15m-4)c_m - \frac{18}{3m-1}\Big) = 27\,s_m. \qquad \square$$

*Proof of Theorem 1.2.* Let $D = \{a \in \mathbb{C} : |a| < \frac{1}{6}\}$. Every term on either side of (5) is holomorphic on a neighbourhood of $\overline{D}$. By Theorem 2.1 both series converge uniformly on $\overline{D}$. By Weierstrass' theorem both sums are holomorphic on $D$, and their Taylor coefficients at $a = 0$ are the sums of the Taylor coefficients of the terms.

On the left, $\frac{1}{(k+c)^2-a^2} = \sum_{j\ge 0} a^{2j}(k+c)^{-2j-2}$, so by (1) the coefficient of $a^{2j}$ is

$$3\Big(\zeta(2j+2,\tfrac{1}{3}) - \zeta(2j+2,\tfrac{2}{3})\Big) = 3^{2j+3}\,L_{-3}(2j+2). \tag{14}$$

On the right, by Proposition 5.2, the coefficients of $a^0$ and $a^2$ are $27\sum_m \tau_m$ and $27\sum_m s_m$. Comparing the coefficients of $a^2$ gives $27\sum_m s_m = 3^5 L_{-3}(4)$, that is, $\sum_m s_m = 9L_{-3}(4)$.

For absolute convergence, Cauchy's estimate on the circle $|a| = \frac{1}{6}$ and (13) give

$$|27\,s_m| = \big|[a^2]\,T_m\big| \le 36 \max_{|a|=1/6} |T_m(a)| \le 36\cdot 3240\,\frac{m^5}{4^{m-1}},$$

so $|s_m| \le \frac{36\cdot 3240\cdot 4}{27}\,m^5 4^{-m} = 17280\,m^5 4^{-m}$. $\square$

## 6. Examples and further consequences

**Corollary 6.1** (Sun's conjecture (2); [3, 2])**.** $\sum_{m\ge 1}\tau_m = L_{-3}(2)$.

*Proof.* Compare the constant terms in the proof of Theorem 1.2: $27\sum_m \tau_m = 3^3 L_{-3}(2)$. $\square$

**Corollary 6.2.** *For every $j \ge 0$,*

$$\sum_{m=1}^{\infty}[a^{2j}]\,T_m(a) = 3^{2j+3}\,L_{-3}(2j+2),$$

*where the series converges absolutely and each $[a^{2j}]T_m(a)$ is an explicit rational number, obtained by expanding the finite products in the proof of Proposition 5.2.*

*Proof.* This is (14) together with the Weierstrass argument above. Absolute convergence follows from Cauchy's estimate $|[a^{2j}]T_m| \le 6^{2j}\cdot 3240\,m^5 4^{1-m}$. $\square$

**Remark 6.3** (Explicit tail bound)**.** The bound $|s_m| \le 17280\,m^5 4^{-m}$ from the proof of Theorem 1.2 controls the tail of Sun's series. For example, $\sum_{m>400}|s_m| < 10^{-224}$.

**Example 6.4.** The first terms of Sun's series are

$$s_1 = \frac{153}{16}, \quad s_2 = -\frac{41777}{32000}, \quad s_3 = \frac{215291781}{878080000}, \quad s_4 = -\frac{29823518547}{584362240000}.$$

For instance, for $m = 1$ we have $H_2^{(2)} = \frac{5}{4}$ and $H_0^{(2)} = 0$, so

$$s_1 = \frac{11 \cdot \frac{45}{4} - 9}{1 \cdot 2^2 \cdot 3} = \frac{153}{16}.$$

Table 1 shows the partial sums $S_M = \sum_{m \le M} s_m$. Since the terms alternate in sign with ratio tending to $-\frac{1}{4}$, each additional term gains about 0.6 decimal digits.

TABLE 1. Partial sums $S_M$ of Sun's series; $9L_{-3}(4) = 8.460231127894113918219625 0\ldots$

| $M$ | $S_M$ | $S_M - 9L_{-3}(4)$ |
|---|---|---|
| 1 | 9.5625 | 1.10 |
| 2 | 8.25696875 | $-2.03 \times 10^{-1}$ |
| 3 | 8.50215344957179300291 5452 | $4.19 \times 10^{-2}$ |
| 5 | 8.46227323015418879987 2142 | $2.04 \times 10^{-3}$ |
| 10 | 8.46022971319614020927 9009 | $-1.41 \times 10^{-6}$ |
| 20 | 8.46023112789315880305 4567 | $-9.55 \times 10^{-13}$ |
| 30 | 8.46023112789411391821 9660 | $-7.44 \times 10^{-19}$ |

## 7. Computer verification

The proof uses a computer at one point: the polynomial identity (9) in Lemma 3.1. Everything else is proved by hand above. The supplementary package contains three Python programs, run together by `src/verify_all.py`. Each writes a JSON certificate to `certificates/`.

(1) `verify_wz.py` (SymPy). It expands both sides of (9) and checks that the difference is zero. It also checks the WZ relation (8) as an identity of rational functions, starting from the two ratios in the proof of Lemma 3.1. Finally, it confirms that $N(n, k)$ is forced: among all $N$ that are quadratic in $(n, k)$ with constant term affine in $a^2$, there is exactly one solution.

(2) `verify_independent.py` (standard library only, exact rational arithmetic). It recomputes everything from the definitions, without using the first program:
- the 343-point grid proof of (9);
- the WZ relation (8) with $F, G$ evaluated directly from (6)–(7), at 1183 points $(n, k, a)$ with $0 \le n, k \le 12$ and seven rational values of $a$;
- the identities of Lemma 4.1;
- the statement of Proposition 5.2, by exact power-series arithmetic modulo $a^3$, for $1 \le m \le 80$; this checks that $[a^0]T_m = 27\tau_m$, $[a^1]T_m = 0$ and $[a^2]T_m = 27s_m$.

(3) `numerical_checks.py` (mpmath, 130-digit working precision). This check is supplementary and is not used in the proof. It sums 400 terms of Sun's series and adds the tail bound of Remark 6.3. The result agrees with $9L_{-3}(4)$, computed from (1) and Hurwitz zeta values, to within $2 \times 10^{-130}$. The same holds for (2). It also checks (5) at $a = \frac{1}{10}$, $\frac{3}{20}i$, $\frac{1}{10} + \frac{1}{9}i$ and $\frac{1}{6}$.

## 8. Scope, attribution and disclosure

Theorem 1.2 proves Sun's Conjecture 3.9 [7] exactly as stated, with no restriction of its range. The essential identity (Theorem 2.1) and its Markov–WZ pair are due to Kh. Hessami Pilehrood and T. Hessami Pilehrood [3]. Our contribution is to identify the conjecture as the $a^2$-coefficient of their identity, to compute that coefficient (Proposition 5.2), and to give a self-contained proof with explicit estimates and exact checks.

This paper was written by Claude Opus 5.5, an AI system developed by Anthropic. Claude Opus 5.5 found the reduction, wrote the proofs and the verification programs, and checked them by internal review and by the computations of Section 7. The work has not been externally peer reviewed. No part of it has been formalized in a proof assistant.

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